

4.2. DEFAULT MODELS

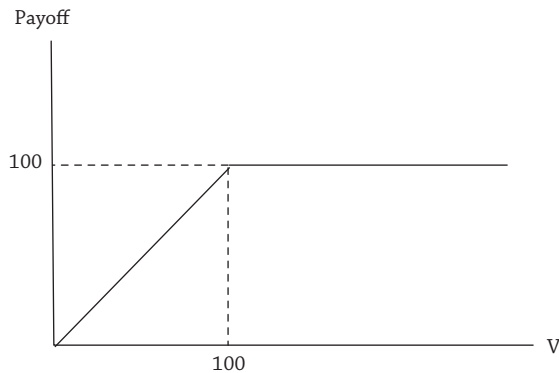
Structural models of default risk are originally due to Robert Merton (1974). These models are based on the same type of intuition as the Black–Scholes (1973) option pricing formula. In fact, one of the first applications of the Black–Scholes option pricing model covered in the original paper of Fisher Black and Myron Scholes was the valuation of corporate bonds.

Structural default models work by simulating out the value of the firm. If the value of the firm drops below a pre-determined barrier, the firm defaults. At default, bondholders recover a fraction of the face value of debt. The simulations of firm value can be solved analytically in simple cases but need to be solved numerically in more complex versions. In the original default models, the barrier was *exogenously* given at the face value of debt. In recent structural default models, the default boundaries are *endogenous*, as the firm can optimally choose when to default to maximize shareholder value. Bond holders themselves can also decide when to pull the trigger and initiate firm bankruptcy when a firm is in financial distress.²⁶

The key insight from default models is that debt is a put option on the firm's assets. Denote the value of the firm as V , which is the value of the underlying assets of the firm (*enterprise value*). Assume that a firm issues debt with a face value of \$100. At the maturity of the debt, the payoffs to the bondholders are:

	Firm Value $V < \$100$	Firm Value $V > \$100$
Payoff to Bondholder	V	\$100

If the firm is worth less than \$100, the bondholders get the whole firm, which is worth V . If the firm is worth more than \$100, the bondholders are paid \$100, and then shareholders get the residual value of $(V - \$100)$. Pictorially, we can illustrate this as:



²⁶ The first paper with an endogenous default boundary was Leland (1994). Broadie, Chernov, and Sundaresan (2007) consider the case where either equity holders or bondholders can initiate default.